

# Kunal Pai

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## EDUCATION

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|---------------------------------------------------------------------------------------------------|--------------------------|
| Ph.D., Computer Science, University of California, Los Angeles                                    | Enrolled: September 2026 |
| M.S., Computer Science, University of California, Davis (GPA: 4.0/4.0)                            | June 2026                |
| B.S., Computer Science & Engineering, University of California, Davis (GPA: 3.8/4.0), with Honors | June 2023                |

## RELEVANT SKILLS

**Languages:** Python, C++, C, JavaScript, Java, Rust

**ML/AI:** TensorFlow, PyTorch, scikit-learn, LLMs, Prompt Engineering, Ollama, Hugging Face, Multi-agent Systems

**Systems & Compilers:** LLVM, Clang, gem5

**Web/Data:** React, Next.js, Django, Flask, MongoDB, pandas, NumPy, Matplotlib

**Tools:** Git, Docker, Unix/Linux, Jupyter

## RESEARCH & PROFESSIONAL EXPERIENCE

**Graduate Student Researcher, DavSec Lab @ UC Davis** April 2025 – Present  
*Research Project* Python, LLMs, C++, Rust, Compiler Analysis

- Built automated C-to-Rust transpilation pipeline using LLMs with 5 prompt variations, benchmarking state-of-the-art models across 746 programs and achieving 70.2% functional accuracy
- Engineered a multi-variant performance-aware transpilation framework that explores alternative Rust implementations in an evolutionary agent setting, yielding up to 3× runtime reduction across benchmarks
- Developed cross-layer analysis framework combining compiler metrics with hardware performance counters to expose semantic and optimization gaps in LLM-translated, syntax-directed and manually-written Rust translations

**Graduate Student Researcher, DECAL Lab @ UC Davis** September 2022 – December 2024  
*Research Project* Python, LLMs, Machine Learning, Code Analysis

- Developed a 4,500-sample dataset for pairwise code-documentation alignment from 200 open-source Python projects, enabling future research in software maintenance
- Engineered a pipeline for measuring calibration and correctness of large language models for code repair, using Defects4J
- Collaborated in validating efficacy of semantic augmentation of language model prompts for code summarization using precision and recall metrics like ROUGE and METEOR

**Teaching Assistant, University of California, Davis** September 2023 – December 2023

- Assisted 180 students in a senior-level Probability & Statistical Modeling class.

## PUBLICATIONS (SELECTED)

**VATS: Exploiting Implicit Authority in Error-Path Injection via Systematic Mutation [SPOTLIGHT]**, Patel, H. & Pai, K., *Second Workshop on Agents in the Wild: Safety, Security, and Beyond, ICML 2026*

**Toward Reproducible and Standardized Computer Architecture Simulation with gem5**, Pai, K., Patel, H., Le, E., et. al., *IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS) 2026*

**NAAMSE: Framework for Evolutionary Security Evaluation of Agents**, Pai, K., Shah, P. & Patel, H., *ICLR 2026 Agents in the Wild: Safety, Security, and Beyond Workshop*

**Implications of Full-System Modeling for Superconducting Architectures**, Pai, K., Samani, M., Nand, A. & Lowe-Power, J., *Workshops of the International Conference for High Performance Computing, Networking, Storage and Analysis (SC Workshops '25)*

**CoDocBench: A Dataset for Code-Documentation Alignment in Software Maintenance**, Pai, K., Devanbu, P. & Ahmed, T., *International Conference on Mining Software Repositories (MSR) 2025: Data and Tool Showcase Track*

**Calibration and Correctness of Language Models for Code**, Spiess, C., Gros, D., Pai, K.S., et. al., *International Conference on Software Engineering (ICSE) 2025*

**Automatic semantic augmentation of language model prompts (for code summarization)**, Ahmed, T., Pai, K.S., Devanbu, P. & Barr, E.T., *International Conference on Software Engineering (ICSE) 2024*

PROJECT EXPERIENCE

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| <b>NAAMSE: Neural Adversarial Agent Mutation-based Security Evaluator</b><br><i>Research Project</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Nov 2025 – Present<br><i>Python, LLMs, Evolutionary Algorithms</i>         |
| <ul style="list-style-type: none"><li>• Won 2nd place (Agent Safety) at the UC Berkeley RDI AgentBeats Competition; framework infrastructure was subsequently forked by Mozilla's Odin team.</li><li>• Designed and implemented a clustering engine to identify semantic attack vectors in LLM-generated adversarial prompts</li><li>• Engineered an attack pipeline that iteratively generates, evaluates, and refines adversarial prompts against target models</li><li>• Benchmarked multiple frontier LLMs within the framework to validate attack effectiveness and refine scoring metrics</li></ul> |                                                                            |
| <b>HASHIRU: Hierarchical, Resource-Aware Multi-Agent Framework</b><br><i>Research Project</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | March 2025 – June 2025<br><i>Python, LLMs, Multi-Agent Systems</i>         |
| <ul style="list-style-type: none"><li>• Designed and deployed a multi-agent architecture enabling dynamic, LLM-driven collaboration across diverse tasks</li><li>• Implemented task decomposition with intelligent agent delegation based on resource cost models and task specialization</li><li>• Engineered autonomous generation of tools and APIs for task execution</li><li>• Developed a robust evaluation framework for agent performance across complex, multi-step tasks</li></ul>                                                                                                              |                                                                            |
| <b>SuperNOVA: Superconducting Graph Accelerator in gem5</b><br><i>Research Project, DArchR Lab @ UC Davis</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | January 2024 – March 2026<br><i>C++, Python, gem5, Hardware Simulation</i> |
| <ul style="list-style-type: none"><li>• Modeled a superconducting graph accelerator and interconnect in gem5, achieving up to 24× speedup and 73× energy efficiency</li><li>• Integrated cryogenic/superconducting cores, caches, and interconnects to evaluate bottlenecks for realistic workloads</li><li>• Mentored undergraduates on modeling, benchmarking, and research writing; one co-authored a ModSim 2024 poster</li></ul>                                                                                                                                                                     |                                                                            |
| <b>gem5 Vision</b><br><i>Resource Discovery Framework</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | January 2023 – June 2023<br><i>Next.js, Python, MongoDB, JSON Schema</i>   |
| <ul style="list-style-type: none"><li>• Accelerated resource discovery 20× for 1,200+ gem5 artifacts by optimizing search and categorization logic</li><li>• Implemented categorization and semantic versioning across 20+ resource types to streamline retrieval</li><li>• Integrated local and remote JSON schemas with MongoDB, improving accessibility for 500+ users</li></ul>                                                                                                                                                                                                                       |                                                                            |

AWARDS AND HONORS

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|--------------------------------------------------------------------------|------------------------------------------------|
| <b>2nd Place (Agent Safety)</b> , UC Berkeley RDI AgentBeats Competition | 2026                                           |
| <b>Dean's List</b> , UC Davis College of Engineering                     | Fall 2019, Fall 2020, Winter 2022, Spring 2022 |
| <b>Provost Award</b> , UC Davis                                          | 2019–2023                                      |

SERVICE

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| <b>Program Committee</b>                                                                                                               |
| <ul style="list-style-type: none"><li>• MSR, Data and Tool Showcase Track (2026)</li></ul>                                             |
| <b>Artifact Evaluation Committee</b>                                                                                                   |
| <ul style="list-style-type: none"><li>• ISSTA (2026)</li><li>• ASE (2026)</li><li>• ISPASS (2026)</li></ul>                            |
| <b>Reviewer</b>                                                                                                                        |
| <ul style="list-style-type: none"><li>• NeurIPS, CODEC-FM Workshop (2026)</li><li>• ICML, Agents in the Wild Workshop (2026)</li></ul> |